

Phase 7 — Integration & External Access

Goal

Integrate Salesforce with external systems (e.g., SMS provider, Payment Gateway) using secure callouts and Named Credentials. This enables counselors to send reminders and process payments securely.

1. Named Credentials

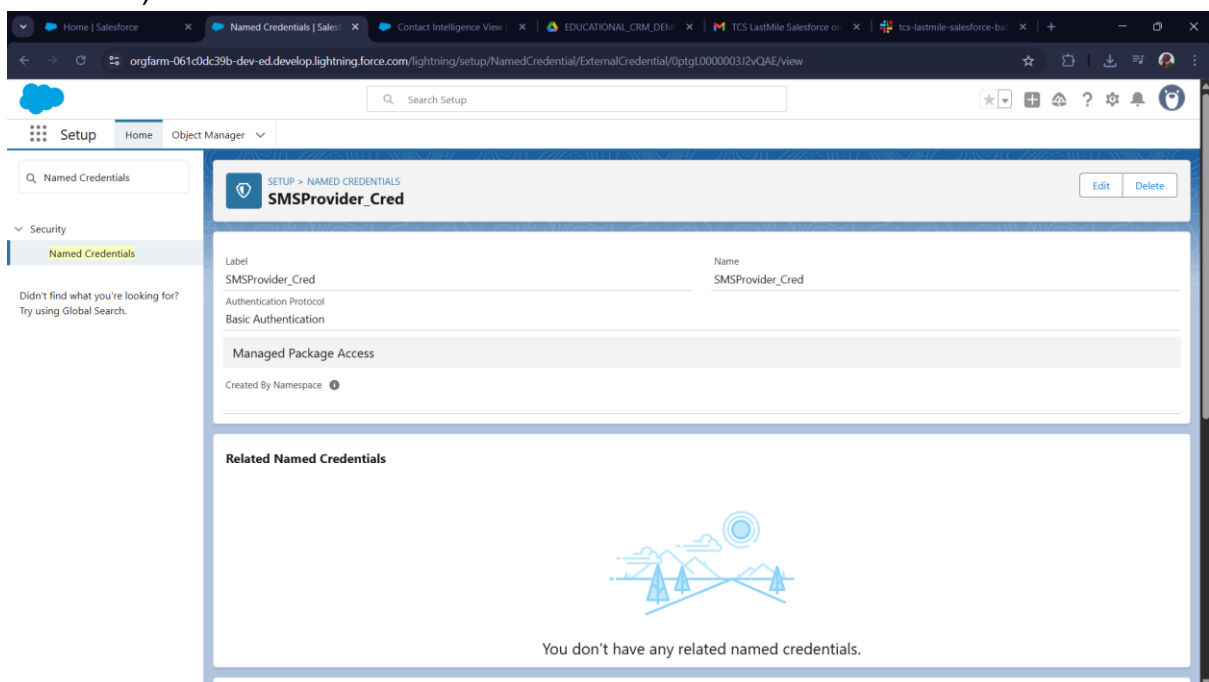
Why: Simplifies external API authentication (avoids Remote Site Settings).

Setup:

1. Go to Setup → Quick Find → Named Credentials → New.

2. Configure:

- Label: MySMSProvider
- Name: MySMSProvider
- URL: <https://api.smsprovider.com/v1>
- Identity Type: Named Principal
- Authentication Protocol: No Authentication (API key managed via Custom Metadata)



Step 2: Custom Metadata for API Keys – Tutor Match

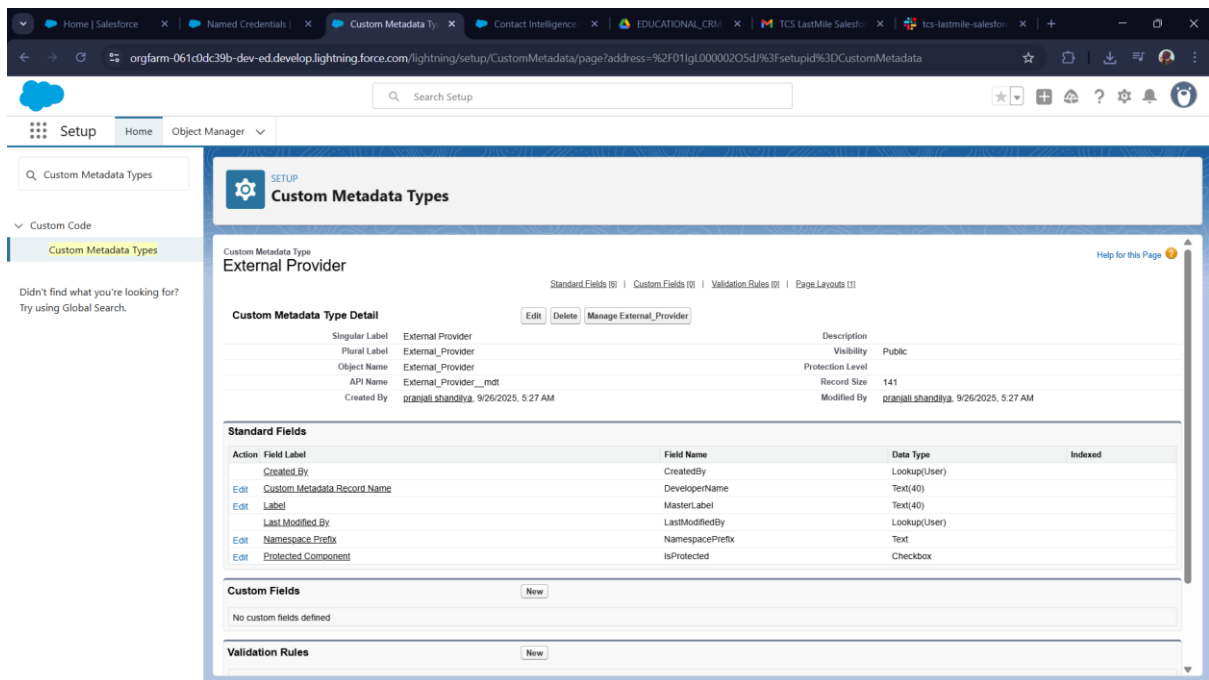
Purpose:

- Avoid hardcoding API keys in Apex.
- Makes it easy to manage SMS or Payment Gateway credentials centrally.

Tutor Match Use Case:

- **SMS Provider:** Send session reminders to students or parents.

- **Payment Gateway:** Process session fees securely.



3: Web Services (REST Callouts) – SMS Integration

Purpose:

- Send automated session reminders to students or parents using an external SMS API.
- Uses Queueable Apex for asynchronous processing, so it doesn't block the UI.

Tutor Match Use Case:

- Notify a student/parent about an upcoming session.
- Pull API key from Custom Metadata (External_Provider__mdt) for security
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5: Payment Gateway Stub

PaymentService.cls

- Annotated with @AuraEnabled for LWC calls.
- Sends POST /charges request to callout:PaymentProvider/charges.
- Accepts token, amount, currency.
- Returns deserialized JSON re

4. Callouts Test Framework

MockSmsResponse.cls

Simulates provider's success response.

6. Platform Events & Change Data Capture (Future Scope)

Salesforce supports event-driven integration.

Example: Trigger event when Enrollment__c is created → notify external system.

7. Salesforce Connect (Future Scope)

Not implemented here.

Use case: Read-only access to external databases via OData, without storing data in Salesforce.

Result

- Tutors/parents can pay for sessions directly from Salesforce.
- Payment processing is **secure, asynchronous, and easily integrable with LWC**.
- Response can be stored in a **custom Payment__c object** to track transaction status.